Problem 1:

$$X = \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \\ x_4(t) \end{bmatrix} \qquad \text{Input} \quad u(t)$$

a) $$\dot{X} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix}$$

$\dot{x}_1 = x_2$ (given) → for non-listed state variables and input, coefficient is 0

$\dot{x}_2 = 9.81 x_3$ (given)

$\dot{x}_3 = x_4$ (given)

$\dot{x}_4 = -2x_4 + u(t)$ (given)

All RHS are in terms of other state variables and $u(t)$, so begin to create matrix

$$\dot{X} = AX(t) + Bu(t)$$

$$\dot{X} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 9.81 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} u$$

A B

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 9.81 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & -2 \end{bmatrix} \qquad B = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

b) Suppose that output is $x_1(t)$

$y(t) = x_1(t)$ → all coefficients for other state variables and input is 0

$$y = Cx(t) + Du(t)$$

$$y = \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + 0\, u(t)$$

↳ only select x_1 → position

$$\Rightarrow \quad C = \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix} \qquad D = [0]$$

Problem 2:

Given Equations:

$$\dot{w} = \frac{NK_m}{J_e} i - \frac{T_L}{J_e} \qquad J_e = J + N^2 J_m$$

$$\frac{di}{dt} = -\frac{R}{L} i + \frac{v}{L} - \frac{NK_m}{L} w$$

$$\dot{\theta} = w$$

output signals: θ, w

Input signal is $v(t)$

Let state vector $= x = \begin{bmatrix} \theta \\ w \\ i \end{bmatrix}$

Let input vector $= u = \begin{bmatrix} v \\ T_L \end{bmatrix}$ ↕ all functions of time, not shown for clarity

includes voltage, disturbance torque

Model:

1. $\dot{X} = Ax + Bu$

$Y = Cx + Du$

$$\dot{X} = \begin{bmatrix} \dot{\theta} \\ \dot{w} \\ \frac{di}{dt} \end{bmatrix}$$

$\dot{\theta} = w$ (given, known)

$\dot{w} = \frac{NK_m}{J_e} i - \frac{1}{J_e} T_L$ (given)

$\frac{di}{dt} = -\frac{NK_m}{L} w - \frac{R}{L} i + \frac{V}{L}$ (given)

All RHS in terms of state vr, input, so solved.

$$\dot{X} = \begin{bmatrix} \dot{\theta} \\ \dot{w} \\ \frac{di}{dt} \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & \frac{NK_m}{J_e} \\ 0 & -\frac{NK_m}{L} & -\frac{R}{L} \end{bmatrix}}_{A} \begin{bmatrix} \theta \\ w \\ i \end{bmatrix} + \underbrace{\begin{bmatrix} 0 & 0 \\ 0 & -\frac{1}{J_e} \\ \frac{1}{L} & 0 \end{bmatrix}}_{B} \begin{bmatrix} V \\ T_L \end{bmatrix}$$

$$y = \begin{bmatrix} \theta \\ w \end{bmatrix}$$

control inputs have no direct effect, $D = 0$ matrix

$$y = \begin{bmatrix} \theta \\ w \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}}_{C} \begin{bmatrix} \theta \\ w \\ i \end{bmatrix} + \underbrace{\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}}_{0} \begin{bmatrix} V \\ T_L \end{bmatrix}$$

Problem 3:

$T_s = 0.1$

States:

$e_d(k)$ → distance error

$e_v(k)$ → velocity error

$a(k)$ → acceleration

Inputs:

$u(k)$ → control acceleration

$a_L(k)$ → acceleration of lead vehicle

1.

$$X(k) = \begin{bmatrix} e_d(k) \\ e_v(k) \\ a(k) \end{bmatrix}$$

$$X(k+1) = \begin{bmatrix} e_d(k+1) \\ e_v(k+1) \\ a(k+1) \end{bmatrix}$$

put in terms of state vectors & input

$$e_d(k+1) = e_d(k) - 0.1 e_v(k) \quad \text{(given)}$$

$$e_v(k+1) = e_v(k) + 0.1 a_2(k) - 0.1 a(k) \quad \text{(given)}$$

$$a(k+1) = \beta a(k) + (1-\beta) u(k) \quad \text{(given)}$$

All RHS is in terms of state var and inputs so proceed

$$X(k+1) = \underbrace{\begin{bmatrix} 1 & -0.1 & 0 \\ 0 & 1 & -0.1 \\ 0 & 0 & \beta \end{bmatrix}}_{A} \begin{bmatrix} e_d(k) \\ e_v(k) \\ a(k) \end{bmatrix} + \underbrace{\begin{bmatrix} 0 & 0 \\ 0 & 0.1 \\ 1-\beta & 0 \end{bmatrix}}_{B} \begin{bmatrix} u(k) \\ a_2(k) \end{bmatrix}$$

$y_1(k) = e_d(k)$

$y_2(k) = e_v(k)$

$\rightarrow$ no direct influence from cont'd inputs $\Rightarrow D = 0$ matrix

$$y(k) = \begin{bmatrix} e_d(k) \\ e_v(k) \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}}_{C} \begin{bmatrix} e_d(k) \\ e_v(k) \\ a(k) \end{bmatrix} + \underbrace{\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}}_{D} \begin{bmatrix} u(k) \\ a_2(k) \end{bmatrix}$$